# VANSHIKA NIMWAL

**SOFTWARE DEVELOPER**

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## PROFESSIONAL SUMMARY

Software Developer with hands-on experience building cross-platform mobile applications, secure backend systems, and immersive XR experiences using Flutter, Java 21, Spring Boot, and Unity. Strong foundation in data structures, object-oriented programming, and software architecture, with practical experience delivering production-ready applications, integrating REST APIs, and developing concurrency-safe, real-time systems, including a double-entry wallet ledger that stays balanced under **100** concurrent transfers, a blockchain-anchored credential verifier with **6** independent checks, and an algorithm visualizer covering **20** algorithms with a live web demo. Passionate about performance optimization, clean code practices, and creating scalable, user-centric digital experiences.

## SKILLS

**Backend & APIs:** Java 21, Spring Boot, REST API Integration, JSON Handling, API Testing, Idempotency & Concurrency Control
**Mobile Development:** Flutter, Dart, Material UI, Cupertino Widgets, Firebase Authentication, Firestore, Figma
**XR / Game Development:** Unity, C#, XR Interaction Toolkit, AR Foundation, 3D Environment Design, Blender
**Programming Languages:** Java, C#, Dart, Python, C++, JavaScript, HTML/CSS
**Software Development:** Object-Oriented Programming (OOP), Data Structures & Algorithms, JavaFX, Multithreading (Basics)
**Databases:** MySQL, PostgreSQL, Firebase Firestore, Redis
**Tools & Platforms:** Android Studio, Unity Editor, IntelliJ IDEA, VS Code, Git, GitHub, Docker, Postman, Vuforia

## EXPERIENCE

**XR Developer Intern - Riyalverse Technology Pvt Ltd** **Oct 2025 – Mar 2026**

- Built immersive AR/VR applications using Unity and C#, using AR Foundation and Vuforia for augmented reality experiences.
- Implemented XR interactions and user controls using XR Interaction Toolkit, making virtual environments responsive and intuitive for users.
- Optimized 3D environments, lighting, and assets (modelled in Blender) for real-time performance, keeping scenes smooth on target devices.

**Software Developer Intern - Alphaaid Healthcare Pvt Ltd** **May 2025 – Oct 2025**

- Developed and maintained healthcare and mobile software applications used by real users, fixing issues and shipping updates.
- Designed modular backend and application components improving performance and scalability, making new features easier to add and test.
- Integrated REST APIs, Firebase services (Authentication, Firestore), and database systems across applications.

**Backend Developer - Capital Bull Wave**

- Built backend APIs for a gaming website: auth, SMS OTP, and player profiles, with **1** mobile number per account and rate limits (about **10** OTP sends per **15** minutes per device) so repeat login/OTP spam is blocked.
- Designed wallet and reward APIs so balances change only on the server. Concurrent clicks and retries use locking and one-time references, keeping **duplicate credits at 0%** when the same request is sent twice.
- Implemented game settlement and admin APIs (bets, live players, account stop/ban). Parallel actions are handled per player, and a single bet cannot exceed **10,000** credits, so two overlapping requests cannot spend the same balance.

**Freelance Flutter Developer - Ferrero Rocher**

- Delivered an internal asset management mobile application in Flutter with real-time data synchronization between users and the backend.
- Integrated MySQL and Firebase backend services with Flutter frontend, keeping data consistent across the application.
- Managed full lifecycle development including architecture, testing, and deployment, from initial design through release.

## PROJECTS

**CertiChain – Blockchain Credential Verification | Java 21, Spring Boot, PostgreSQL, Docker** GitHub

- Built a credential platform where an institution issues **3** credential types (degree, transcript, bonafide) as signed PDFs, and a public page runs **6** separate checks: issuer record, Ed25519 signature, SHA-256 file hash, revocation, expiry, and chain anchor.
- Implemented **3** role-based views: an admin console, a student wallet that shows only credentials issued to that email with expiring share links, and a verifier desk for checking a credential ID.
- Anchored each certificate's SHA-256 fingerprint on a persistent local blockchain (state survives restarts), with Sepolia support using the institution's own RPC URL and signing key.

**PayFlow – Wallet & Payments Simulation | Java 21, Spring Boot, PostgreSQL, Redis, Flutter** GitHub

- Built a wallet simulation with a double-entry ledger in integer paise so every transfer, **2%** shop fee, **1%** cashback line, and refund sums to zero and user balances cannot go negative.
- Prevented double payments and deadlocks with idempotency keys (verified under a **50**-way replay), account locks in ascending id order (**1,000** A-to-B and B-to-A transfers each way), and a top-up flow that calls the mock bank only after the database lock is released.
- Shipped a Flutter client for register, send, top-up, and history with a recipient check before money moves, backed by integration tests for **100** concurrent transfers, idempotent retries, refunds, and webhook recovery.

**AlgoVerse – Algorithm Visualizer | Java 21, JavaFX, HTML, CSS, JavaScript** Live Demo | GitHub

- Built an interactive visualizer for **20** algorithms (**12** sorting, **3** search, **5** pathfinding) plus linked lists and BSTs, with play/pause, step-through, and speed control.
- Added side-by-side sort comparison on the same array, highlighted pseudocode, and an info panel covering complexity and when to use each algorithm.
- Shipped a no-install browser demo (plain HTML, CSS, JS) on GitHub Pages plus a matching Java 21 + JavaFX desktop app, so the same animations run on the web and locally.

## EDUCATION

**Sharda University** **Aug 2023 – Present**

Bachelor of Technology in Computer Science, Greater Noida India

## CERTIFICATION & ACHIEVEMENTS

- Java Foundation and Advanced Certification – Oracle
- Android Studio: Complete Guide of Android App Development – Udemy
- Dart Mastery: Become a Dart Master From Zero to Hero – Udemy
- Flutter & Firebase Chat App: Master Flutter and Firebase – Udemy